

Methods of Proof



This Lecture

Now we have learnt the basics in logic.

We are going to apply the logical rules in proving mathematical theorems.

We will illustrate different proof techniques including:

- Direct proof
- Contrapositive
- Proof by contradiction
- Proof by cases

Basic Definitions

An integer n is an **even** number
if there exists an integer k such that $n = 2k$.

An integer n is an **odd** number
if there exists an integer k such that $n = 2k+1$.

Proving an Implication

Goal: If P, then Q. (P implies Q)

Method 1: Write assume P, then show that Q logically follows.

The sum of two even numbers is even.

Proof

$$\begin{aligned}x &= 2m, y = 2n \\x+y &= 2m+2n \\&= 2(m+n)\end{aligned}$$

Direct Proofs

The product of two odd numbers is odd.

Proof

$$\begin{aligned}x &= 2m+1, y = 2n+1 \\xy &= (2m+1)(2n+1) \\&= 4mn + 2m + 2n + 1 \\&= 2(2mn+m+n) + 1.\end{aligned}$$

If m and n are perfect square, then $m+n+2\sqrt{mn}$ is a perfect square.

Proof

$$\begin{aligned}m &= a^2 \text{ and } n = b^2 \text{ for some integers } a \text{ and } b \\ \text{Then } m + n + 2\sqrt{mn} &= a^2 + b^2 + 2ab \\ &= (a + b)^2 \\ \text{So } m + n + 2\sqrt{mn} &\text{ is a perfect square.}\end{aligned}$$

Divisibility

a "divides" b ($a|b$):

$$b = ak \text{ for some integer } k$$

$$5|15 \text{ because } 15 = 3 \cdot 5$$

$$n|0 \text{ because } 0 = n \cdot 0$$

$$1|n \text{ because } n = 1 \cdot n$$

$$n|n \text{ because } n = n \cdot 1$$

A number $p > 1$ with no positive integer divisors other than 1 and itself is called a **prime**. Every other number greater than 1 is called **composite**.

2, 3, 5, 7, 11, and 13 are prime,
4, 6, 8, and 9 are composite.

Simple Divisibility Facts

1. If $a \mid b$, then $a \mid bc$ for all c .
2. If $a \mid b$ and $b \mid c$, then $a \mid c$.
3. If $a \mid b$ and $a \mid c$, then $a \mid sb + tc$ for all s and t .
4. For all $c \neq 0$, $a \mid b$ if and only if $ca \mid cb$.

Proof of (1)

$$a \mid b$$

$$\Rightarrow b = ak$$

$$\Rightarrow bc = ack$$

$$\Rightarrow bc = a(ck)$$

$$\Rightarrow a \mid bc$$

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Proof of (2)

$$a \mid b \Rightarrow b = ak_1$$

$$b \mid c \Rightarrow c = bk_2$$

$$\Rightarrow c = ak_1k_2$$

$$\Rightarrow a \mid c$$

a "divides" b ($a \mid b$):

$$b = ak \text{ for some integer } k$$

Simple Divisibility Facts

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4. For all $c \neq 0$, $a \mid b$ if and only if $ca \mid cb$.

Proof of (3)

$$a \mid b \Rightarrow b = ak_1$$

$$a \mid c \Rightarrow c = ak_2$$

$$sb + tc$$

$$= sak_1 + tak_2$$

$$= a(sk_1 + tk_2)$$

$$\Rightarrow a \mid (sb + tc)$$

a "divides" b ($a \mid b$):

$$b = ak \text{ for some integer } k$$

Simple Divisibility Facts

1. If $a \mid b$, then $a \mid bc$ for all c .
2. If $a \mid b$ and $b \mid c$, then $a \mid c$.
3. If $a \mid b$ and $a \mid c$, then $a \mid sb + tc$ for all s and t .
4. For all $c \neq 0$, $a \mid b$ if and only if $ca \mid cb$.

a "divides" b ($a \mid b$):

$b = ak$ for
some integer k

Proof of (4) (only if part:)
[For all $c \neq 0$, if $a \mid b$ then $ca \mid cb$.]

$$\begin{aligned} b &= ak \\ \Rightarrow cb &= cak \\ \Rightarrow ca &\mid cb \end{aligned}$$

Proof of (4) (if part:)
[For all $c \neq 0$, if $ca \mid cb$ then $a \mid b$.]

$$\begin{aligned} cb &= cak \\ \Rightarrow b &= ak \\ \Rightarrow a &\mid b \end{aligned}$$

This Lecture

- Direct proof
- **Contrapositive**
- Proof by contradiction
- Proof by cases

Proving an Implication

Goal: If P , then Q . (P implies Q)

Method 1: Write assume P , then show that Q logically follows.

Claim: If r is irrational, then $\sqrt{r}$ is irrational.

How to begin with?

What if I prove "If $\sqrt{r}$ is rational, then r is rational", is it equivalent?

Yes, this is equivalent, because it is the **contrapositive** of the statement, so proving "if P , then Q " is equivalent to proving "if not Q , then not P ".

Rational Number

R is **rational** $\Leftrightarrow$ there are integers a and b such that

$$\begin{array}{c} \text{numerator} \rightarrow \\ r = \frac{a}{b} \quad \text{and } b \neq 0. \\ \text{denominator} \rightarrow \end{array}$$

Is 0.281 a rational number?

Yes, 281/1000

Is 0 a rational number?

Yes, 0/1

If m and n are non-zero integers, is $(m+n)/mn$ a rational number?

Yes

Is the sum of two rational numbers a rational number?

Yes, $a/b + c/d = (ad+bc)/bd$

Is $x=0.12121212\dots$ a rational number?

Note that $100x - x = 12$, and so $x = 12/99$.

Proving the Contrapositive

Goal: If P , then Q . (P implies Q)

Method 2: Prove the *contrapositive*, i.e. prove "not Q implies not P ".

Claim: If r is irrational, then $\sqrt{r}$ is irrational.

Proof:

We shall prove the contrapositive -

"if $\sqrt{r}$ is rational, then r is rational."

Since $\sqrt{r}$ is rational, $\sqrt{r} = a/b$ for some integers a, b .

So $r = a^2/b^2$. Since a, b are integers, a^2, b^2 are integers.

Therefore, r is rational. $\square$ Q.E.D.

(Q.E.D.) "which was to be demonstrated", or "quite easily done". ☺

Proving an “if and only if”

Goal: Prove that two statements P and Q are “logically equivalent”, that is, one holds if and only if the other holds.

Example: For an integer n , n is even if and only if n^2 is even.

Method 1: Prove P implies Q and Q implies P .

Method 1': Prove P implies Q and not P implies not Q .

Proof the Contrapositive

For an integer n , n is even if and only if n^2 is even.

Method 1: Prove P implies Q and Q implies P .

Statement: If n is even, then n^2 is even

Proof: $n = 2k$

$$n^2 = 4k^2$$

Statement: If n^2 is even, then n is even

Proof: $n^2 = 2k$

$$n = \sqrt{2k}$$

??

Proof the Contrapositive

For an integer n , n is even if and only if n^2 is even.

Method 1': Prove P implies Q and not P implies not Q .

Statement: If n^2 is even, then n is even

Contrapositive: If n is odd, then n^2 is odd.

Proof (the contrapositive):

Since n is an odd number, $n = 2k+1$ for some integer k .

$$\begin{aligned}\text{So } n^2 &= (2k+1)^2 \\ &= (2k)^2 + 2(2k) + 1 = 2(2k^2 + 2k) + 1\end{aligned}$$

So n^2 is an odd number.

This Lecture

- Direct proof
- Contrapositive
- Proof by contradiction
- Proof by cases

Proof by Contradiction

$$\frac{\bar{P} \rightarrow \mathbf{F}}{P}$$

To prove P , you prove that not P would lead to ridiculous result,
and so P must be true.

Proof by Contradiction

Theorem: $\sqrt{2}$ is irrational.

Proof (by contradiction): [sketch of the proof]

- Suppose $\sqrt{2}$ was rational.
- Choose m, n integers **without common prime factors** (always possible)
such that $\sqrt{2} = \frac{m}{n}$
- Show that m and n are both even, thus having a common factor 2,
a contradiction!

Proof by Contradiction

m, n integers **without common prime factors** (always possible)

Theorem: $\sqrt{2}$ is irrational.

$$\sqrt{2} = \frac{m}{n}$$

Proof (by contradiction):

Want to prove both m and n are even.

$$\sqrt{2} = \frac{m}{n}$$

$$\sqrt{2}n = m$$

$$2n^2 = m^2$$



so m is even.

so can assume $m = 2l$

$$m^2 = 4l^2$$

$$2n^2 = 4l^2$$

$$n^2 = 2l^2$$

so n is even.

For an integer n , n is even if and only if n^2 is even.

Infinitude of the Primes

Theorem. There are infinitely many prime numbers.

Proof (by contradiction): [Proof sketch]

Assume there are only finitely many primes.

Let $p_1, p_2, \dots, p_k$ be all the primes.

We will construct a number N so that N is not divisible by any p_i .

By our assumption, it means that N is not divisible by any prime number.

On the other hand, we show that any integer must be divided by *some* prime.

It leads to a contradiction, and therefore the assumption must be false.

So there must be infinitely many primes.

CONTRADICTION

Divisibility by a Prime

Theorem. Any integer $n > 1$ is divisible by a prime number.

- Let n be an integer.
- If n is a prime number, then we are done.
- Otherwise, $n = ab$, both are smaller than n .
- If a or b is a prime number, then we are done.
- Otherwise, $a = cd$, both are smaller than a .
- If c or d is a prime number, then we are done.
- Otherwise, repeat this argument, since the numbers are getting smaller and smaller, this will eventually stop and we have found a prime factor of n .

Idea of induction.

Infinitude of the Primes

Theorem. There are infinitely many prime numbers.

Proof (by contradiction):

Let $p_1, p_2, \dots, p_N$ be all the primes.

Consider an integer: $p_1 p_2 \dots p_N + 1$.

Claim: if a prime p divides a , then p does not divide $a+1$.

Proof (by contradiction):

$a = cp$ for some integer c

$a+1 = dp$ for some integer d

$\Rightarrow 1 = (d-c)p$, contradiction because $p \geq 2$.

So none of $p_1, p_2, \dots, p_N$ can divide $p_1 p_2 \dots p_N + 1$, a contradiction.

This Lecture

- Direct proof
- Contrapositive
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- Proof by cases

Proof by Cases

$$p \vee q$$

$$p \rightarrow r$$

$$q \rightarrow r$$

$$\therefore r$$

e.g. want to prove a nonzero number always has a positive square.

x is positive or x is negative

if x is positive, then $x^2 > 0$.

if x is negative, then $x^2 > 0$.

$$\therefore x^2 > 0.$$

The Square of an Odd Integer

$$\forall \text{ odd } n, \exists m, n^2 = 8m + 1?$$

Idea 0: find counterexample.

$$3^2 = 9 = 8+1, \quad 5^2 = 25 = 3 \times 8 + 1 \quad \dots \quad 131^2 = 17161 = 2145 \times 8 + 1, \dots$$

Idea 1: prove that $n^2 - 1$ is divisible by 8.

$$n^2 - 1 = (n-1)(n+1) = ??...$$

Idea 2: consider $(2k+1)^2$

$$(2k+1)^2 = 4k^2 + 4k + 1$$

If k is even, then both k^2 and k are even

If k is odd, then both k^2 and k are odd, and so $k^2 + k$ even

$$\text{So, either way } 4k^2 + 4k + 1 = 4(k^2 + k) + 1 = 4(2m) + 1 = 8m + 1$$

Rational vs Irrational

Question: If a and b are irrational, can a^b be rational??

We (only) know that $\sqrt{2}$ is irrational, what about $\sqrt{2}^{\sqrt{2}}$?

Case 1: $\sqrt{2}^{\sqrt{2}}$ is rational

Then we are done, $a=\sqrt{2}$, $b=\sqrt{2}$.

Case 2: $\sqrt{2}^{\sqrt{2}}$ is irrational

Then $(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^2 = 2$, a rational number

So $a=\sqrt{2}^{\sqrt{2}}$, $b=\sqrt{2}$ will do.

So in either case there are a, b irrational and a^b be rational.

We don't (need to) know which case is true!

Summary

We have learnt different techniques to prove mathematical statements.

- Direct proof
- Contrapositive
- Proof by contradiction
- Proof by cases

Next time we will focus on a very important technique, proof by induction.